Activity 6: Website Flow Sankey Design

Course: DSA0601 – Data Handling and Visualization

Slot: C

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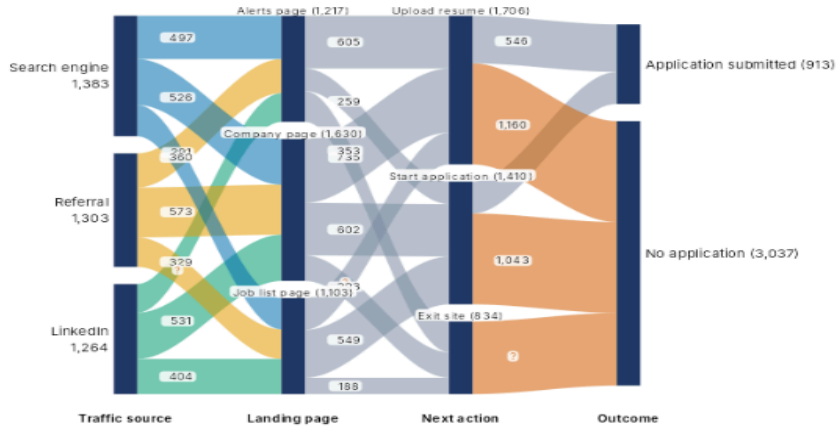
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DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot C

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Scenario: The analytics team of a job portal tracked 3,950 visits last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of visits on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Job list page to Upload resume; Referral to Job list page; Exit site to No application. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Application submitted as a percentage of all visits; (ii) the largest single flow into "No application" or "Exit site"; (iii) the percentage of "Job list page" visits that go on to "Start application". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 200 visits. Compute the height in cm of each landing-page node (Job list page, Company page, Alerts page). [2]

(e) Write R code: a links data frame for the first four flows leaving "LinkedIn" and "Search engine" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

Scenario

The analytics team of a job portal tracked 3,950 visits last month through four stages: traffic source, landing page, next action, and outcome. The Sankey diagram below shows the number of visits along different flows. Three flows have hidden values marked

Given data from the diagram

1: Traffic sourceStage

2: Landing pageStage

3: Next actionStage

4: Outcome

Part A — Questions and answers

(a) Find the three hidden flows. Show your working and state the rule used. (4 marks)

Rule: Conservation of flow — the total entering a node must equal the total leaving that node.

1. Job list page → Upload resume

$$1103 = 298 + 185 + 620$$

Hidden flow = $1103 - 298 - 185 = 620$**.**

Answer: 620 visits.

2. Referral → Job list page

$$1303 = 401 + 573 + 329$$

Hidden flow = $1303 - 401 - 573 = 329$**.**

Answer: 329 visits.

3. Exit site → No application

$$834 = 363 + 471$$

Hidden flow = $834 - 363 = 471$**.**

Answer: 471 visits.

(b) Compute the following.

(i) Overall conversion rate

\[\frac{913}{3950}\times100=\boxed{23.11\%} \]

Answer: 23.11% of all visits resulted in an application submission.

(ii) Largest single flow into "No application" or "Exit site"

From the diagram, the largest clearly labelled flow into No application is Start application → No application, with 1,024 visits.

Answer: 1,024 visits.

(iii) Percentage of Job list page visits that go on to Start application

Using the diagram's visible Job list page → Start application flow of 479 visits:

\[\frac{479}{1103}\times100=\boxed{43.43\%} \]

Answer: 43.43%.

(c) Design the final Sankey diagram. Give four design decisions and justify each.

- ❖ Node order: Arrange Traffic source → Landing page → Next action → Outcome from left to right to show the process clearly.
- ❖ Colour rule: Use different colours for traffic sources and actions to distinguish the flows.
- ❖ Labels: Display node names, visit totals, and important flow values to make the chart easy to understand.
- ❖ Highlighting: Highlight Application submitted in a contrasting colour to emphasize successful conversions.
- ❖

(d) Compute the height of each landing-page node.

Given: 1 cm of node height = 200 visits.

Formula:

$$Heightincm = \frac{Visits}{200}$$

(e) Write R code.

1. Links data frame for the first four flows leaving LinkedIn and Search engine

```
links <- data.frame(
  source = c(
    "Search engine", "Search engine",
    "Search engine", "LinkedIn"
  ),
  target = c(
    "Alerts page", "Company page",
    "Job list page", "Alerts page"
  ),
  value = c(497, 526, 360, 401)
)
```

2. Nodes data frame and Sankey plot

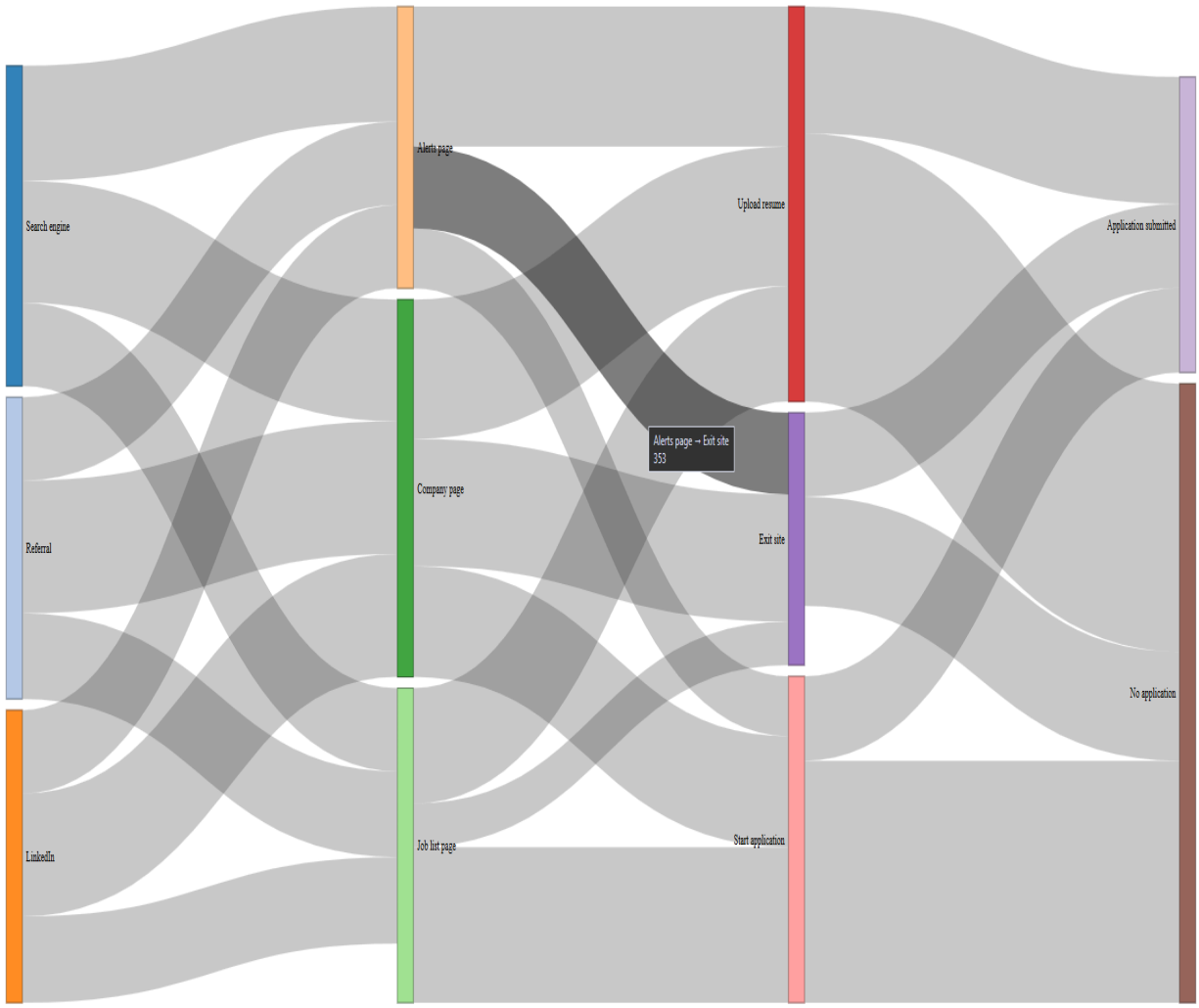
```
library(networkD3)


nodes <- data.frame(
  name = c(
    "Search engine", "Referral", "LinkedIn",
    "Alerts page", "Company page", "Job list page",
```

```
    "Upload resume", "Start application", "Exit site",
    "Application submitted", "No application"
  )
)

links$source <- match(links$source, nodes$name) - 1
links$target <- match(links$target, nodes$name) - 1

sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 12,
  nodeWidth = 25
)
```



Note: The code above illustrates a partial Sankey using the four listed flows. To reproduce the complete diagram, add all source-to-landing, landing-to-action, and action-to-outcome links.

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each.

- ❖ Too many nodes and crossing flows: The diagram becomes cluttered and difficult to follow. Alternative: A grouped bar chart.
- ❖ Comparing exact values across many flows: Similar ribbon widths are difficult to compare accurately. Alternative: A table or bar chart.

Result

The Sankey diagram represents the flow of 3,950 job-portal visits across four stages. It helps identify traffic sources, popular landing pages, user actions, and application outcomes. The overall application conversion rate is 23.11%.